the Sun couldn’t hold onto its planets; they would fly away into interstellar space.

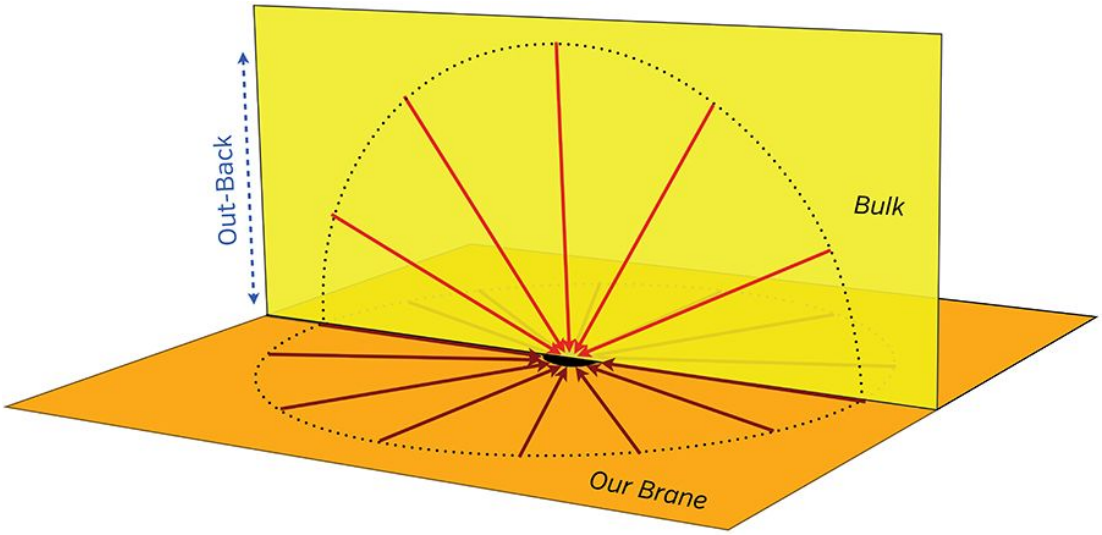


Fig. 23.2. Gravitational force lines spread radially into the bulk, if the bulk is not warped. The dotted circles are solely to guide your eyes. *[Patterned on a figure in Lisa Randall’s* Warped Passages *(Randall 2006).]*

But they don’t fly away. And their measured motions reveal unequivocally that the Sun’s gravity weakens as the inverse square of the distance. The conclusion is inescapable: if there *is* a bulk, it must be warped in some manner that prevents gravity from spreading into the fifth dimension, the *out-back* dimension.

Is Out-Back Curled Up?

If the bulk’s out-back dimension were *curled up* into a tight roll, then gravity could not spread far into the bulk, and the inverse square law would be restored.